

MUSKOKA, ON, Canada

WMO: 715320

Lat: 44.973N

Lon: 79.304W

Elev: 282

StdP: 97.98

Time Zone: -5.00 (NAE)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.9	-23.3	-31.1	0.2	-26.0	-27.4	0.3	-22.7	9.3	-7.5	8.4	-8.2	1.4	320	0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	29.8	21.3	28.1	20.3	26.7	19.5	22.7	27.5	21.8	26.0	20.9	24.5	4.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.3	16.5	24.6	20.5	15.7	23.8	19.6	14.8	23.1	68.6	27.7	65.2	26.2	61.8	24.6	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.4	6.6	-33.1	32.4	3.5	1.2	-35.6	33.3	-37.7	34.0	-39.7	34.7	-42.2	35.6		
(5)			-33.2	24.5	3.5	1.0	-35.7	25.3	-37.8	25.9	-39.7	26.4	-42.2	27.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.8	-9.3	-8.6	-3.2	4.7	11.8	16.7	19.5	18.5	14.6	7.8	1.2	-5.4	(6)
(7)		DBStd	11.45	7.48	6.51	6.59	5.13	4.63	3.65	3.03	3.02	4.25	4.66	5.24	6.28	(7)
(8)		HDD10.0	2588	599	521	411	173	33	1	0	0	8	100	264	477	(8)
(9)		HDD18.3	4731	858	754	667	410	210	73	23	34	127	326	513	735	(9)
(10)		CDD10.0	1045	0	0	2	14	88	203	294	264	146	33	1	0	(10)
(11)		CDD18.3	147	0	0	0	1	7	25	59	41	14	1	0	0	(11)
(12)		CDH23.3	1467	0	0	1	11	103	276	568	370	131	7	0	0	(12)
(13)		CDH26.7	325	0	0	0	1	22	64	133	81	24	0	0	0	(13)
(14)	Wind	WSAvg	3.0	3.2	3.2	3.3	3.4	3.0	2.7	2.5	2.4	2.5	3.0	3.2	3.3	(14)
(15)	Precipitation	PrecAvg	1102	96	69	74	78	90	78	95	111	101	117	115	(15)	
(16)		PrecMax	1580	140	145	130	124	175	213	178	178	194	164	197	202	(16)
(17)		PrecMin	857	27	16	14	30	20	32	8	26	48	39	64	52	(17)
(18)		PrecStd	152	28	36	25	27	32	45	40	34	40	37	36	37	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.7	7.2	16.6	25.0	29.3	30.7	31.8	31.5	29.5	24.3	16.1	10.0	(19)
(20)			MCWB	7.1	5.6	11.4	15.4	19.8	21.8	22.1	21.7	21.8	17.8	11.6	8.6	(20)
(21)		2%	DB	4.8	4.4	10.9	20.3	26.1	28.6	29.7	28.8	26.8	20.8	13.1	6.6	(21)
(22)			MCWB	4.0	3.1	7.1	12.1	17.5	20.5	21.4	20.7	20.2	15.4	10.7	5.6	(22)
(23)		5%	DB	2.5	2.2	8.0	16.7	23.6	26.6	28.1	27.0	24.5	18.0	11.1	4.3	(23)
(24)			MCWB	1.7	0.9	5.0	10.2	15.4	19.0	20.4	20.2	18.7	14.9	8.6	3.5	(24)
(25)		10%	DB	0.7	0.6	5.6	13.7	20.9	24.5	26.6	25.3	22.5	15.5	8.9	2.4	(25)
(26)			MCWB	0.0	-0.4	3.2	8.7	14.1	17.7	19.3	18.9	17.9	13.0	7.4	1.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.4	6.0	13.4	16.9	21.1	23.4	24.1	23.4	22.7	19.5	13.2	9.2	(27)
(28)			MCD B	7.5	6.5	16.1	21.8	26.8	28.6	29.2	27.5	26.9	22.1	14.4	9.9	(28)
(29)		2%	WB	4.1	3.2	8.1	14.1	19.3	22.0	22.8	22.3	21.6	16.9	11.1	5.8	(29)
(30)			MCD B	4.5	4.2	9.8	18.7	24.3	26.9	27.6	26.6	24.9	19.7	12.9	6.5	(30)
(31)		5%	WB	1.8	1.1	5.4	11.5	17.4	20.5	21.7	21.4	20.1	15.1	9.0	3.6	(31)
(32)			MCD B	2.4	1.9	7.4	15.1	21.2	24.2	25.9	25.0	22.9	17.4	10.5	4.2	(32)
(33)		10%	WB	0.1	-0.3	3.4	9.2	15.6	19.2	20.8	20.2	18.7	13.3	7.4	1.7	(33)
(34)			MCD B	0.6	0.6	5.5	13.0	19.0	22.7	24.5	23.5	21.6	15.0	8.9	2.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.9	10.8	10.6	11.3	12.8	12.1	12.0	11.9	9.6	8.2	8.0	(35)	
(36)			MCD BR	8.1	10.1	12.0	17.4	16.8	15.2	14.4	14.1	14.2	13.6	11.9	7.5	(36)
(37)			MCWBR	7.7	9.1	8.6	10.3	8.4	7.6	6.8	6.9	7.9	8.6	9.3	7.1	(37)
(38)		5% WB	MCD BR	7.8	9.3	10.5	14.0	13.0	12.4	11.6	11.2	11.6	11.7	10.4	7.1	(38)
(39)			MCWBR	7.6	8.9	8.4	9.8	7.7	7.2	6.3	6.5	7.2	8.3	9.0	7.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.287	0.292	0.313	0.354	0.379	0.404	0.406	0.398	0.370	0.343	0.308	0.282	(40)	
(41)		taud	2.373	2.342	2.410	2.380	2.354	2.301	2.317	2.332	2.406	2.471	2.516	2.439	(41)	
(42)		Ebn at Noon	829	894	924	907	890	865	858	852	850	827	806	801	(42)	
(43)		Edh at Noon	75	93	100	113	120	128	124	118	101	83	66	64	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.34	2.22	3.60	4.49	5.29	5.77	5.83	5.03	3.95	2.31	1.36	0.99	(44)	
(45)		RadStd	0.16	0.28	0.37	0.42	0.53	0.45	0.41	0.28	0.38	0.25	0.18	0.12	(45)	

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page